BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 157 · Timed Paper 2 Practice

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. In a file-size calculation, should the final answer include units?

Q2. Q = (A AND B) OR C. A=1, B=1, C=0. What is Q?
A. 1
B. 0
C. 2
D. 10

Q3. “A translator is needed only to change the program's comments into another human language.” Correct this. Explain why.

Q4. Merge the sorted lists [2, 7] and [4, 6] into ascending order. Give the values in order. Why not just join the lists?

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I Do · Read new knowledge

Key vocabulary

Paper 2 timing — The OCR J277/02 paper has 80 marks in 90 minutes.

Code tracing — Write a reliable value table before selecting the output.

Program design — Plan input, processing, selection, iteration and output.

Error spotting — Check indexes, loop bounds, equality tests and branching.

New learning

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Worked example

Trace count=0; for i=1 to 5; if i MOD 2=0 then count=count+1. Output count.

1. Loop through i=1,2,3,4,5.

2. Only i=2 and 4 satisfy even test.

3. Count progresses 0,1,1,2,2.

4. Final output is 2.

We Do · Practise with guidance

Guided 1

Design a three-line correction to a failed validation check using OR.

Worked method: Check endpoints.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Count the 1 bits in a five-bit binary value using iteration and selection.

```
array bits = [1,0,1,1,0]
ones = 0
for i = 0 to 4
    if bits[i] == 1 then
        ones = ones + 1
    endif
next i
print(ones)
```

Predict the count of 1 bits.

Change bits[1] from 0 to 1; count again.

Convert the original unsigned binary 10110 into denary using its place values.

Question 1

State what an accumulator stores. [2 marks]

Question 2

Write a valid condition for 0≤score≤100. [2 marks]

Question 3

Explain one common index error in zero-based arrays. [3 marks]

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Exit Ticket · Answer before checking

A trace table helps follow loop values and decisions.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Yes; show a calculation and label the final unit, such as bits or bytes.

Q2. A — 1

Q3. A CPU runs machine instructions. High-level source code must be translated before or as it runs.

Q4. (a) [2, 4, 6, 7]. Keep taking the smaller next item. Just joining the lists would put 7 before 4.

I Do / We Do pages · Guided practice

Q1. Check endpoints.

You Do pages · Code First

Expected result: The original five-bit number 10110 has three 1 bits, so output is 3. Changing index 1 to 1 gives four 1 bits (11110). The original binary value is 16+4+2 = 22 denary; counting set bits and converting place values are different operations.

You Do pages · Independent questions

Q1. A running total updated as a program executes.

Q2. score >= 0 AND score <= 100.

Q3. Using length instead of length−1 as final index can access outside array bounds.

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding